

LAKENHEATH, United Kingdom

WMO: 035830

Lat: 52.417N

Lon: 0.567E

Elev: 33

StdP: 14.68

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 35034

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	26.8	21.1	15.9	29.0	23.4	17.7	30.3	31.2	50.3	27.8	50.2	3.8	230	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	82.2	65.8	78.3	64.3	75.0	63.0	68.4	77.2	66.4	74.5	64.8	71.9	10.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.9	95.7	70.5	63.9	88.9	68.5	62.3	84.0	67.4	32.6	76.6	31.0	74.2	29.8	71.5	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.7	22.9	20.5	DB	18.8	88.7	4.0	3.9	15.9	91.5	13.6	93.8	11.3	96.0	8.4	98.8	(4)
			WB	18.2	71.6	4.2	2.7	15.2	73.6	12.7	75.2	10.4	76.7	7.4	78.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	40.0	41.0	44.3	48.7	54.5	59.5	64.2	63.4	58.2	52.6	45.4	41.3
	DBStd	10.23	6.15	6.27	5.40	6.02	5.72	5.00	4.77	4.85	4.79	5.80	6.22	7.00
	HDD50	1357	313	256	189	95	22	1	0	0	1	39	163	278
	HDD65	5167	775	673	642	489	328	176	74	90	211	386	589	734
	CDD50	1778	3	4	12	56	161	287	441	415	247	119	24	9
	CDD65	112	0	0	0	0	2	12	50	40	7	1	0	0
	CDH74	923	0	0	0	6	30	122	423	288	45	9	0	0
	CDH80	206	0	0	0	0	1	23	100	73	8	1	0	0
Wind	WSAvg	9.6	10.7	11.0	10.6	9.3	9.6	9.1	8.6	8.8	8.5	9.0	9.5	10.6
Precipitation	PrecAvg	24.7	2.0	1.7	1.5	1.6	1.9	2.2	2.3	2.6	2.0	2.4	2.5	2.0
	PrecMax	31.2	3.7	3.1	3.5	4.8	4.6	6.2	4.5	5.9	4.4	4.5	4.3	3.9
	PrecMin	16.4	0.4	0.3	0.2	0.0	0.5	0.6	0.8	0.3	0.4	0.4	0.8	0.8
	PrecStd	4.2	0.9	0.7	0.8	1.2	1.1	1.4	1.0	1.4	1.1	1.0	0.9	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.7	59.2	63.0	73.6	77.3	83.8	88.4	86.5	79.9	71.9	61.2	58.0
		MCWB	53.5	53.1	53.8	58.2	61.9	67.0	69.1	67.7	65.5	62.9	58.2	54.0
	2%	DB	54.0	54.0	59.3	67.0	73.5	77.2	82.2	81.6	73.5	66.4	59.1	55.7
		MCWB	51.5	49.9	53.1	56.1	60.7	64.3	65.7	65.9	63.6	61.0	55.7	53.2
	5%	DB	51.8	52.1	56.8	62.7	69.6	73.4	78.6	76.6	69.9	63.5	57.0	53.8
		MCWB	49.4	48.8	51.3	54.0	59.3	62.0	64.4	64.1	61.5	59.3	54.0	51.6
	10%	DB	49.8	50.2	53.7	59.3	65.9	70.0	75.1	73.2	67.3	61.0	55.0	51.8
		MCWB	47.6	47.5	49.0	52.2	57.1	60.6	63.4	62.3	60.0	57.6	52.3	49.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.6	53.7	56.9	60.2	65.8	69.6	71.3	69.9	69.1	65.0	59.3	55.5
		MCDB	55.5	56.0	60.5	70.4	72.3	79.5	80.2	80.9	75.1	69.3	60.9	56.5
	2%	WB	52.4	51.3	54.1	57.4	62.8	66.1	68.4	67.4	65.6	62.3	56.6	53.8
		MCDB	54.1	53.5	57.9	64.2	69.1	72.8	77.8	76.7	70.9	65.4	58.3	55.5
	5%	WB	50.0	49.7	52.0	55.5	60.4	63.7	66.4	65.7	63.6	60.2	54.5	52.1
		MCDB	51.6	51.8	55.6	61.4	67.5	70.2	75.3	73.9	68.2	62.9	56.3	53.6
	10%	WB	47.8	47.9	49.9	53.1	58.1	61.7	64.5	64.0	61.5	58.5	52.5	50.1
		MCDB	49.3	50.1	53.3	58.7	64.6	68.5	72.3	70.9	65.3	60.8	54.4	51.8
Mean Daily Temperature Range	5% DB	MDBR	10.1	11.8	14.2	17.1	17.2	16.7	17.8	17.1	16.2	14.0	11.4	10.0
		MCDBR	11.3	13.3	18.2	22.0	23.5	22.9	23.9	22.0	19.4	14.8	10.8	10.2
		MCWBR	10.5	10.4	12.0	12.9	12.7	11.7	10.6	9.7	11.1	9.9	9.0	9.5
	5% WB	MCDBR	11.6	12.4	16.2	19.2	20.5	19.4	20.5	19.2	16.8	12.9	11.2	10.4
MCWBR		11.3	10.6	12.0	12.3	12.1	11.4	10.8	9.6	11.2	9.8	10.1	10.0	
Clear-Sky Solar Irradiance	taub	0.327	0.344	0.379	0.403	0.407	0.411	0.410	0.399	0.377	0.358	0.331	0.317	
	taud	2.354	2.369	2.264	2.212	2.246	2.256	2.271	2.323	2.381	2.417	2.401	2.341	
	Ebn at Noon	209	242	254	263	268	267	265	262	253	234	207	190	
	Edh at Noon	19	25	34	40	41	41	40	36	30	24	18	17	
All-Sky Solar Radiation	RadAvg	232	447	799	1258	1524	1621	1549	1304	979	570	288	184	
	RadStd	22	66	89	131	115	126	161	93	81	51	30	23	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page